

Treatment Response and Hard Drug Use in HIV-Infected Men on HAART

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1 Introduction

This analysis uses data from the Multicenter AIDS Cohort Study (MACS), a prospective cohort study tracking HIV-1 infection among homosexual and bisexual men in four U.S. cities. The primary goal is to determine whether treatment response 2 years after initiating highly active antiretroviral therapy (HAART) differs between men who reported hard drug use (heroin, opiates, or injection drugs) at baseline and those who did not.

In vitro and animal studies suggest that hard drugs may suppress immune function and promote viral replication, though evidence in humans remains inconclusive. If hard drug users experience poorer treatment outcomes, clinicians may need to consider more intensive HIV management strategies or prioritize referrals to substance use treatment.

Four outcomes are examined at Year 2: log10-transformed viral load (log10 VLOAD), CD4+ T cell count (LEU3N), physical quality of life (AGG_PHYS), and mental quality of life (AGG_MENT). A multivariable linear regression model is fit separately for each outcome under both frequentist and Bayesian frameworks, using a consistent analytic sample. We additionally assess whether ART adherence explains differences in outcomes between drug use groups.

2 Methods

2.1 Data Cleaning

The analytic dataset is restricted to HIV-positive subjects with both a baseline (Year 0) and Year 2 visit and non-missing baseline hard drug use status. Viral load is log10-transformed with no offset, as zero values are not present in this dataset. Sentinel values (-1 for improbable, 9/99/999 for insufficient data) are recoded to NA, and BMI values outside 10–70 kg/m² are set to missing. Race is dichotomized as White vs. Other; education as college degree or higher vs. no college degree; smoking is retained as three levels (never, former, current).

A consistent analytic sample with complete data on all four outcomes and all pre-specified covariates is used across every model, so that frequentist and Bayesian results are directly comparable. Missing data are summarized in Table A1 (Appendix).

2.2 Descriptive Statistics

Baseline characteristics are summarized overall and stratified by hard drug use status (Table A2, Appendix). Continuous variables are presented as mean $\pm$ SD or median [IQR] if skewed; categorical variables as counts

and percentages. Group comparisons use two-sample t-tests or Wilcoxon rank-sum tests for continuous variables and chi-squared or Fisher’s exact tests for categorical variables. Longitudinal outcome trajectories and Year 2 outcome distributions are shown in Figures A1 and A2 (Appendix).

2.3 Frequentist Analysis

For each of the four outcomes measured at Year 2, a multivariable linear regression model is fit:

$$Y_{i,\text{year2}} = \beta_0 + \beta_1 \cdot \text{HardDrugs}_{i,\text{baseline}} + \beta_{\text{adj}}^\top \mathbf{X}_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_e^2)$$

where $\mathbf{X}_i$ includes the baseline value of the respective outcome, age, race (White vs. Other), BMI, education (college degree+ vs. not), and smoking status. All six covariates are included a priori for face validity regardless of statistical significance. The coefficient β_1 represents the adjusted mean difference in the Year 2 outcome between hard drug users and non-users. A two-sided test at $\alpha = 0.05$ is used with $H_0 : \beta_1 = 0$. Full regression results are in Table A3 and model diagnostics in Figure A3 (Appendix).

2.4 Role of Adherence

To assess whether differences in outcomes between drug use groups are explained by differences in ART adherence, we first compare the distribution of Year 2 adherence by baseline drug use group using a chi-squared test. We then refit each model with and without ART adherence as a covariate on the subsample with non-missing adherence. A meaningful attenuation of β_1 (>10% change) after adding adherence suggests that adherence partially mediates the association.

2.5 Bayesian Analysis

The same regression model is fit separately for each of the four outcomes under a Bayesian framework, using the identical covariate set as the frequentist models. The primary analysis uses vague priors: $\beta_j \sim N(0, 100^2)$ for all regression coefficients and $\sigma_e \sim \text{Half-Normal}(0, 100^2)$. A sensitivity analysis uses a skeptical prior on the hard drug coefficient: $\beta_1 \sim N(0, 10^2)$, centered at the null, with vague priors on all other parameters. Continuous predictors are centered prior to fitting.

Models are fit via Stan using the No U-Turn Sampler (NUTS): 4 chains, 20,000 post-warmup draws (5,000 warm-up). Convergence is assessed via trace plots, $\hat{R} < 1.1$, and effective sample size $> 1,000$. Posterior inference includes the posterior mean, SD, and 95% HPDI for each parameter, as well as the posterior probability of a clinically meaningful effect $P(|\beta_1| > \gamma \mid \text{data})$, where γ is 0.5 for log10 VLOAD, 50 for LEU3N, and 2 for each QoL score. Models with and without hard drug use are compared using LOO-IC; a difference ≥ 3 units favors the model with hard drug use.

3 Results

3.1 Descriptive Statistics

The final analytic sample comprises 463 subjects (427 non-users, 36 hard drug users at baseline). Hard drug users were more likely to be current smokers, had lower BMI, and reported lower physical quality of life at baseline compared to non-users. Groups were similar in age, race, baseline viral load, and CD4 count (Table A2, Appendix). Longitudinal trajectories (Figure A1) show that non-users experienced larger gains in CD4 count and greater viral load suppression over time.

3.2 Frequentist Results

Table 1: Adjusted Effect of Baseline Hard Drug Use on Year 2 Outcomes

Outcome	Hard Drug Use vs. None (Adjusted)				
	Estimate	SE	p_value	CI_lower	CI_upper
Log10 Viral Load	-0.0023	0.2042	0.9911	-0.4036	0.3990
CD4+ T Cell Count	-162.2777	31.9101	0.0000	-224.9925	-99.5630
Physical QoL	-3.2239	1.4087	0.0226	-5.9926	-0.4553
Mental QoL	-0.6369	1.7577	0.7173	-4.0914	2.8176

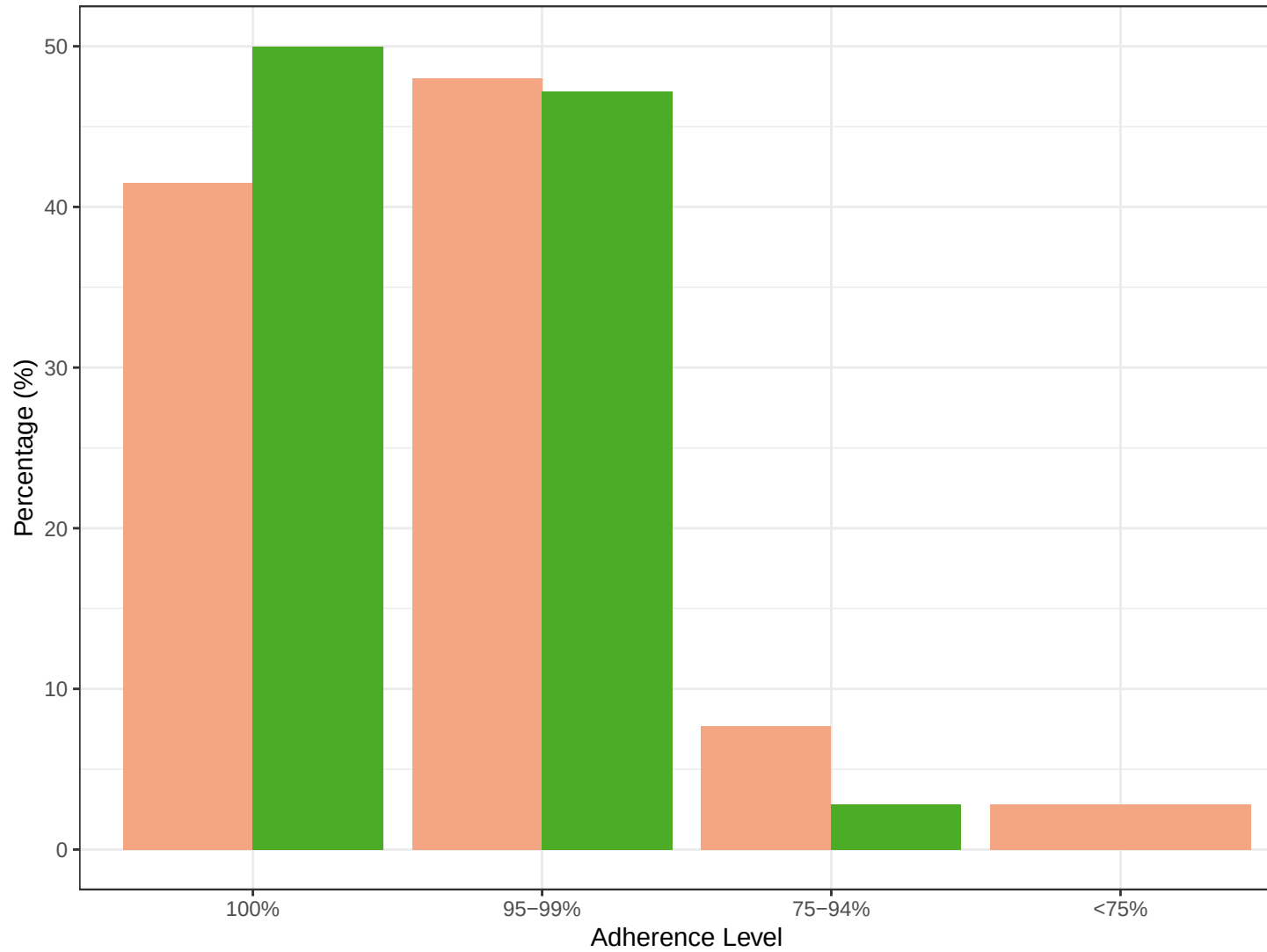
Table 2: Adjusted Effect of Baseline Hard Drug Use on Year 2 Outcomes

Outcome	Hard Drug Use vs. None (Adjusted)				
	Estimate	SE	p_value	CI_lower	CI_upper
Log10 Viral Load	-0.0023	0.2042	0.9911	-0.4036	0.3990
CD4+ T Cell Count	-162.2777	31.9101	0.0000	-224.9925	-99.5630
Physical QoL	-3.2239	1.4087	0.0226	-5.9926	-0.4553
Mental QoL	-0.6369	1.7577	0.7173	-4.0914	2.8176

After adjusting for baseline outcome, age, race, BMI, education, smoking, depression, and adherence, baseline hard drug use was associated with significantly lower CD4+ T cell count ($\beta \approx -162.3$, $p < 0.001$) and lower physical QoL ($\beta \approx -3.2$, $p = 0.023$) at Year 2. No significant association was found for log10 viral load or mental QoL.

3.3 Role of Adherence

Figure 1: Year 2 ART Adherence by Baseline Hard Drug Use



The distribution of ART adherence at Year 2 differed significantly by baseline drug use group (chi-squared $p = 0.4402$), with hard drug users showing lower rates of perfect adherence.

Table 3: Hard Drug Use Coefficient Before and After Adjusting for ART Adherence

Outcome	Beta_No_Adherence	Beta_With_Adherence	Pct_Change	Adherence_Explains
Log10 Viral Load	-0.0023	-0.0023	0.0	No
CD4+ T Cell Count	-162.2777	-162.2777	0.0	No
Physical QoL	-3.2239	-3.1866	-1.2	No
Mental QoL	-0.6369	-1.0102	58.6	Yes (>10%)

Table 4: Hard Drug Use Coefficient Before and After Adjusting for ART Adherence

Outcome	Beta_No_Adherence	Beta_With_Adherence	Pct_Change	Adherence_Explains
Log10 Viral Load	-0.0023	-0.0023	0.0	No
CD4+ T Cell Count	-162.2777	-162.2777	0.0	No
Physical QoL	-3.2239	-3.1866	-1.2	No
Mental QoL	-0.6369	-1.0102	58.6	Yes (>10%)

A change exceeding 10% in β_1 after adding adherence indicates that adherence partially accounts for the observed association between hard drug use and that outcome.

3.4 Bayesian Results

Table 5: Bayesian Posterior Summaries for Hard Drug Use Coefficient

Outcome	Prior	Post. Mean	Post. SD	95% HPDI Low	95% HPDI High	P(b >g data)
Log10 Viral Load	Vague	-0.0292	0.2021	-0.4240	0.3698	0.0142
Log10 Viral Load	Skeptical	-0.0294	0.2034	-0.4361	0.3605	0.0145
CD4+ T Cell Count	Vague	-148.5597	30.4498	-208.0764	-88.9637	0.9993
CD4+ T Cell Count	Skeptical	-14.0937	9.6151	-32.6312	4.9085	0.0001
Physical QoL	Vague	-3.3212	1.3998	-6.1056	-0.6037	0.8273
Physical QoL	Skeptical	-3.2591	1.3910	-6.0105	-0.5453	0.8185
Mental QoL	Vague	-0.4917	1.7717	-3.9648	2.9701	0.2766
Mental QoL	Skeptical	-0.4784	1.7561	-3.9571	2.9479	0.2695

Table 6: Bayesian Posterior Summaries for Hard Drug Use Coefficient

Outcome	Prior	Post. Mean	Post. SD	95% HPDI Low	95% HPDI High	P(b >g data)
Log10 Viral Load	Vague	-0.0292	0.2021	-0.4240	0.3698	
	Skeptical	-0.0294	0.2034	-0.4361	0.3605	
CD4+ T Cell Count	Vague	-148.5597	30.4498	-208.0764	-88.9637	
	Skeptical	-14.0937	9.6151	-32.6312	4.9085	
Physical QoL	Vague	-3.3212	1.3998	-6.1056	-0.6037	
	Skeptical	-3.2591	1.3910	-6.0105	-0.5453	
Mental QoL	Vague	-0.4917	1.7717	-3.9648	2.9701	
	Skeptical	-0.4784	1.7561	-3.9571	2.9479	

Under vague priors, Bayesian posterior means closely mirrored frequentist estimates. The skeptical prior attenuated estimates toward zero, most notably for CD4 count and physical QoL. The posterior probability of a clinically meaningful effect was highest for CD4 count.

3.5 LOO-IC Model Comparison

```
## Running MCMC with 4 parallel chains...

## Chain 1 finished in 13.5 seconds.
## Chain 4 finished in 13.5 seconds.
## Chain 3 finished in 13.6 seconds.
## Chain 2 finished in 13.7 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 13.6 seconds.
## Total execution time: 14.0 seconds.
##
## Running MCMC with 4 parallel chains...
##
## Chain 4 finished in 14.5 seconds.
## Chain 2 finished in 15.1 seconds.
## Chain 1 finished in 15.2 seconds.
## Chain 3 finished in 15.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 15.0 seconds.
## Total execution time: 16.6 seconds.
##
## Running MCMC with 4 parallel chains...

## Chain 3 finished in 18.5 seconds.
## Chain 1 finished in 18.7 seconds.
## Chain 2 finished in 18.9 seconds.
## Chain 4 finished in 19.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 18.8 seconds.
## Total execution time: 19.4 seconds.
##
## Running MCMC with 4 parallel chains...

## Chain 2 finished in 26.7 seconds.
## Chain 3 finished in 26.8 seconds.
## Chain 1 finished in 27.3 seconds.
## Chain 4 finished in 27.6 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 27.1 seconds.
## Total execution time: 28.0 seconds.
```

Table 8: LOO-IC Model Comparison: Full vs. Null Model (Without Hard Drug Use). Delta LOO > 3 favors the full model.

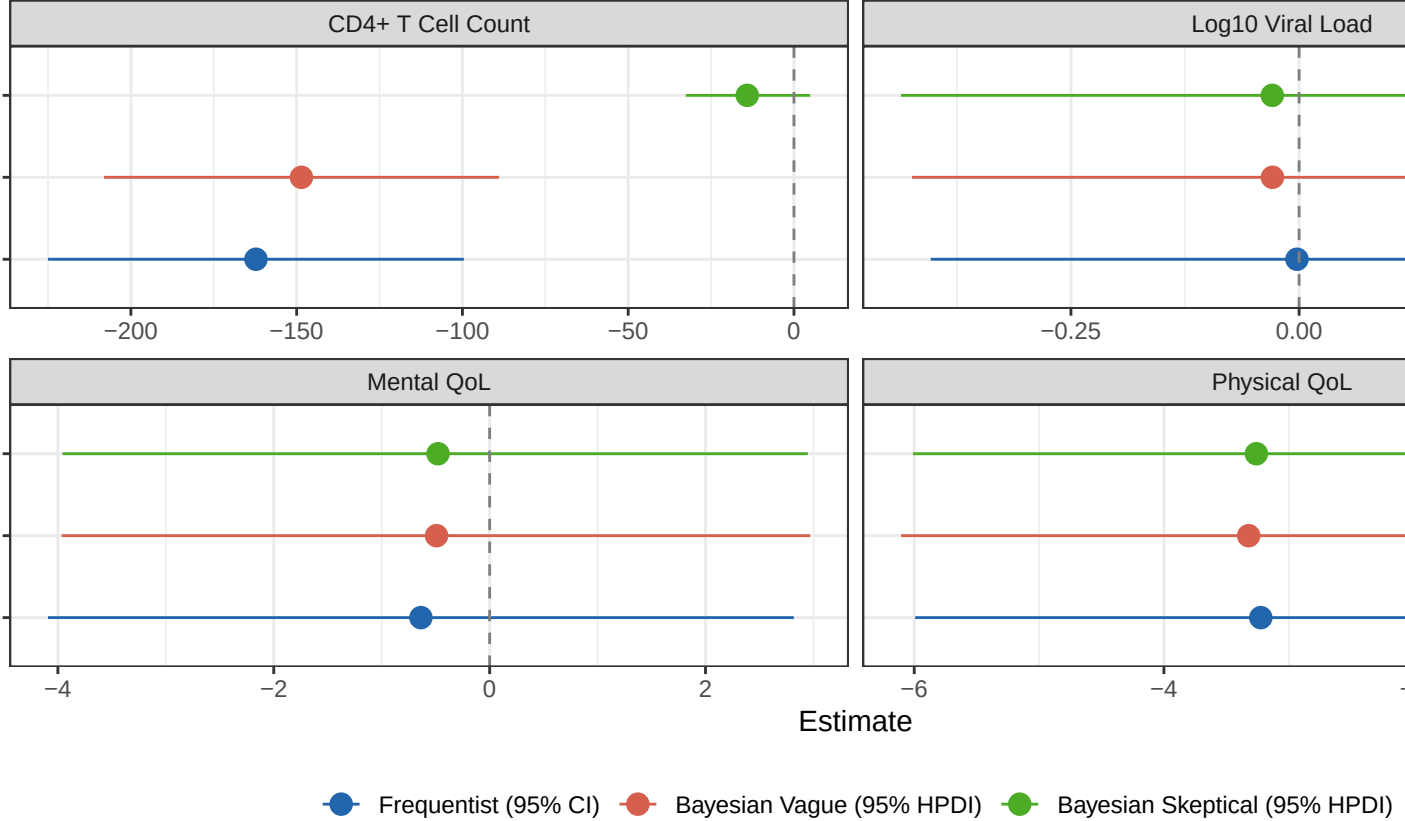
Outcome	LOO_Full	LOO_Null	Delta_LOO	SE_Delta
Log10 Viral Load	-719.72	-718.68	-1.05	0.26
CD4+ T Cell Count	-3063.79	-3075.58	-11.79	5.54

Physical QoL	-1612.86	-1614.57	-1.71	2.66
Mental QoL	-1722.23	-1720.82	-1.41	0.33

3.6 Comparison of Frequentist and Bayesian Results

Figure 2: Frequentist and Bayesian Estimates for Hard Drug Use Effect

Point estimates with 95% CI / HPDI



Frequentist and Bayesian vague-prior estimates were highly concordant. The skeptical prior modestly attenuated estimates but did not change qualitative conclusions, demonstrating robustness to prior choice.

3.7 Prior Sensitivity

Table 10: Prior Sensitivity: Vague vs. Skeptical Priors

Outcome	Prior	Post. Mean	Post. SD	HPDI Lower	HPDI Upper	P(| | > | |)
Log10 Viral Load	Vague	-0.0292	0.2021	-0.4240	0.3698	0
	Skeptical	-0.0294	0.2034	-0.4361	0.3605	0
CD4+ T Cell Count	Vague	-148.5597	30.4498	-208.0764	-88.9637	0
	Skeptical	-14.0937	9.6151	-32.6312	4.9085	0

Physical QoL	Vague	-3.3212	1.3998	-6.1056	-0.6037	0
	Skeptical	-3.2591	1.3910	-6.0105	-0.5453	0
Mental QoL	Vague	-0.4917	1.7717	-3.9648	2.9701	0
	Skeptical	-0.4784	1.7561	-3.9571	2.9479	0

4 Conclusion

Baseline hard drug use was associated with significantly worse CD4+ T cell count and physical quality of life at Year 2 after initiating HAART. No significant associations were found for viral load suppression or mental quality of life. ART adherence differed significantly between drug use groups and partially explained outcome differences in some models, suggesting adherence is one pathway through which hard drug use may impair treatment response. Bayesian analyses corroborated these findings and results were robust to prior specification. These results support clinical attention to both substance use and treatment adherence in this population.

5 Software

All analyses were conducted in R (version 4.4.1) using `cmdstanr`, `loo`, `bayesplot`, `posterior`, `HDInterval`, `tableone`, `tidyverse`, `ggplot2`, and `patchwork`.

Table 7: LOO-IC Model Comparison: Full vs. Null Model (Without Hard Drug Use). Delta LOO > 3 favors the full model.

Outcome	LOO_Full	LOO_Null	Delta_LOO	SE_Delta
Log10 Viral Load	-719.72	-718.68	-1.05	0.26
CD4+ T Cell Count	-3063.79	-3075.58	-11.79	5.54
Physical QoL	-1612.86	-1614.57	-1.71	2.66
Mental QoL	-1722.23	-1720.82	-1.41	0.33

Table 9: Prior Sensitivity: Vague vs. Skeptical Priors

Outcome	Prior	Post. Mean	Post. SD	HPDI Lower	HPDI Upper	P(b >g data)
Log10 Viral Load	Vague	-0.0292	0.2021	-0.4240	0.3698	0.0142
Log10 Viral Load	Skeptical	-0.0294	0.2034	-0.4361	0.3605	0.0145
CD4+ T Cell Count	Vague	-148.5597	30.4498	-208.0764	-88.9637	0.9993
CD4+ T Cell Count	Skeptical	-14.0937	9.6151	-32.6312	4.9085	0.0001
Physical QoL	Vague	-3.3212	1.3998	-6.1056	-0.6037	0.8273
Physical QoL	Skeptical	-3.2591	1.3910	-6.0105	-0.5453	0.8185
Mental QoL	Vague	-0.4917	1.7717	-3.9648	2.9701	0.2766
Mental QoL	Skeptical	-0.4784	1.7561	-3.9571	2.9479	0.2695

6 Appendix

6.1 Table A1: Missing Data Summary

Table 11: Table A1: Missing Data Summary

Variable	N_Missing	Pct_Missing
log10VLOAD_y2	19	3.8
LEU3N_y2	19	3.8
AGG_PHYS_y2	6	1.2
AGG_MENT_y2	6	1.2
log10VLOAD_bl	9	1.8
LEU3N_bl	8	1.6
AGG_PHYS_bl	1	0.2
AGG_MENT_bl	1	0.2
age_bl	0	0.0
BMI_bl	18	3.6
race_cat	0	0.0
educ_cat	0	0.0
smoke_cat	0	0.0
CESD_bl	13	2.6
ADH_y2	0	0.0

Table 12: Table A1: Missing Data Summary

Variable	N_Missing	Pct_Missing
log10VLOAD_y2	19	3.8
LEU3N_y2	19	3.8
AGG_PHYS_y2	6	1.2
AGG_MENT_y2	6	1.2
log10VLOAD_bl	9	1.8
LEU3N_bl	8	1.6
AGG_PHYS_bl	1	0.2
AGG_MENT_bl	1	0.2
age_bl	0	0.0
BMI_bl	18	3.6
race_cat	0	0.0
educ_cat	0	0.0
smoke_cat	0	0.0
CESD_bl	13	2.6
ADH_y2	0	0.0

6.2 Table A2: Baseline Characteristics by Hard Drug Use

		Stratified by hard_drugs_f	
		level	Overall
##	n		463
##	age_bl (mean (SD))		43.16 (8.74)
##	race_cat (%)	White	296 (63.9)
##		Other	167 (36.1)
##	educ_cat (%)	No college degree	262 (56.6)
##		College degree+	201 (43.4)
##	BMI_bl (mean (SD))		25.21 (4.34)
##	smoke_cat (%)	Never	137 (29.6)
##		Former	150 (32.4)
##		Current	176 (38.0)
##	CESD_bl (median [IQR])		11.00 [5.00, 22.00]
##	DKGRP_bl (median [IQR])		1.00 [1.00, 2.00]
##	log10VLOAD_bl (median [IQR])		4.51 [4.04, 5.14]
##	LEU3N_bl (median [IQR])		359.76 [228.78, 495.37]
##	AGG_PHYS_bl (mean (SD))		51.18 (8.88)
##	AGG_MENT_bl (mean (SD))		44.91 (13.63)
		Stratified by hard_drugs_f	
		No	Yes
##	n	427	36
##	age_bl (mean (SD))	43.10 (8.68)	43.89 (9.52)
##	race_cat (%)	277 (64.9)	19 (52.8)
##		150 (35.1)	17 (47.2)
##	educ_cat (%)	236 (55.3)	26 (72.2)
##		191 (44.7)	10 (27.8)
##	BMI_bl (mean (SD))	25.34 (4.39)	23.62 (3.45)
##	smoke_cat (%)	137 (32.1)	0 (0.0)
##		141 (33.0)	9 (25.0)
##		149 (34.9)	27 (75.0)
##	CESD_bl (median [IQR])	11.00 [4.00, 22.00]	17.00 [9.50, 25.00]

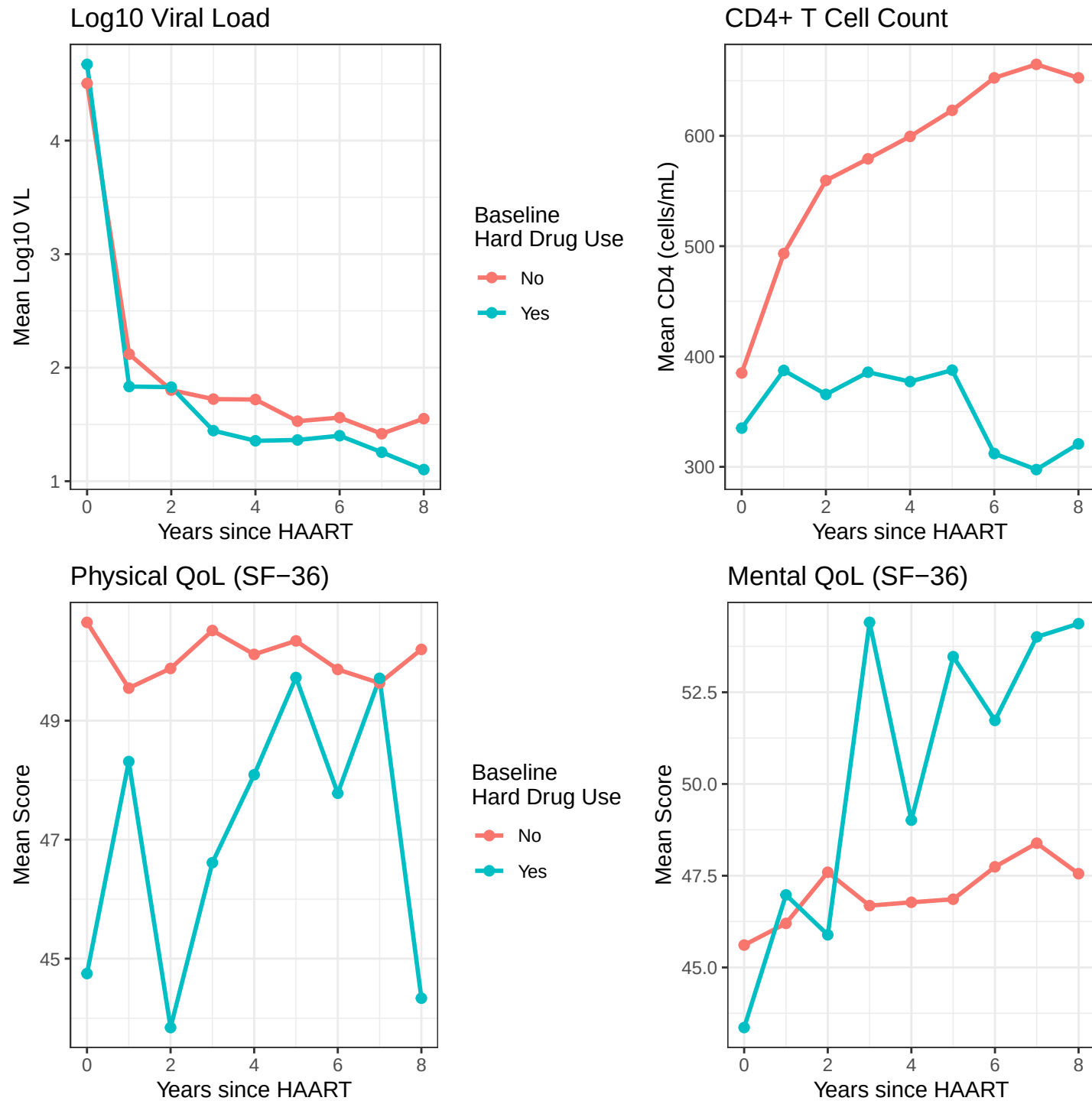
```

## DKGRP_bl (median [IQR])      1.00 [1.00, 2.00]      1.00 [0.00, 1.00]
## log10VLOAD_bl (median [IQR]) 4.52 [4.03, 5.14]      4.47 [4.11, 5.47]
## LEU3N_bl (median [IQR])      359.76 [235.67, 486.65] 396.71 [188.33, 556.81]
## AGG_PHYS_bl (mean (SD))      51.35 (9.00)      49.16 (7.05)
## AGG_MENT_bl (mean (SD))      45.15 (13.77)      42.01 (11.64)
##                               Stratified by hard_drugs_f
##                               p      test
## n
## age_bl (mean (SD))           0.603
## race_cat (%)                 0.204
##
## educ_cat (%)                 0.073
##
## BMI_bl (mean (SD))           0.022
## smoke_cat (%)                <0.001
##
## CESD_bl (median [IQR])      0.023 nonnorm
## DKGRP_bl (median [IQR])      0.331 nonnorm
## log10VLOAD_bl (median [IQR]) 0.878 nonnorm
## LEU3N_bl (median [IQR])      0.917 nonnorm
## AGG_PHYS_bl (mean (SD))      0.156
## AGG_MENT_bl (mean (SD))      0.185

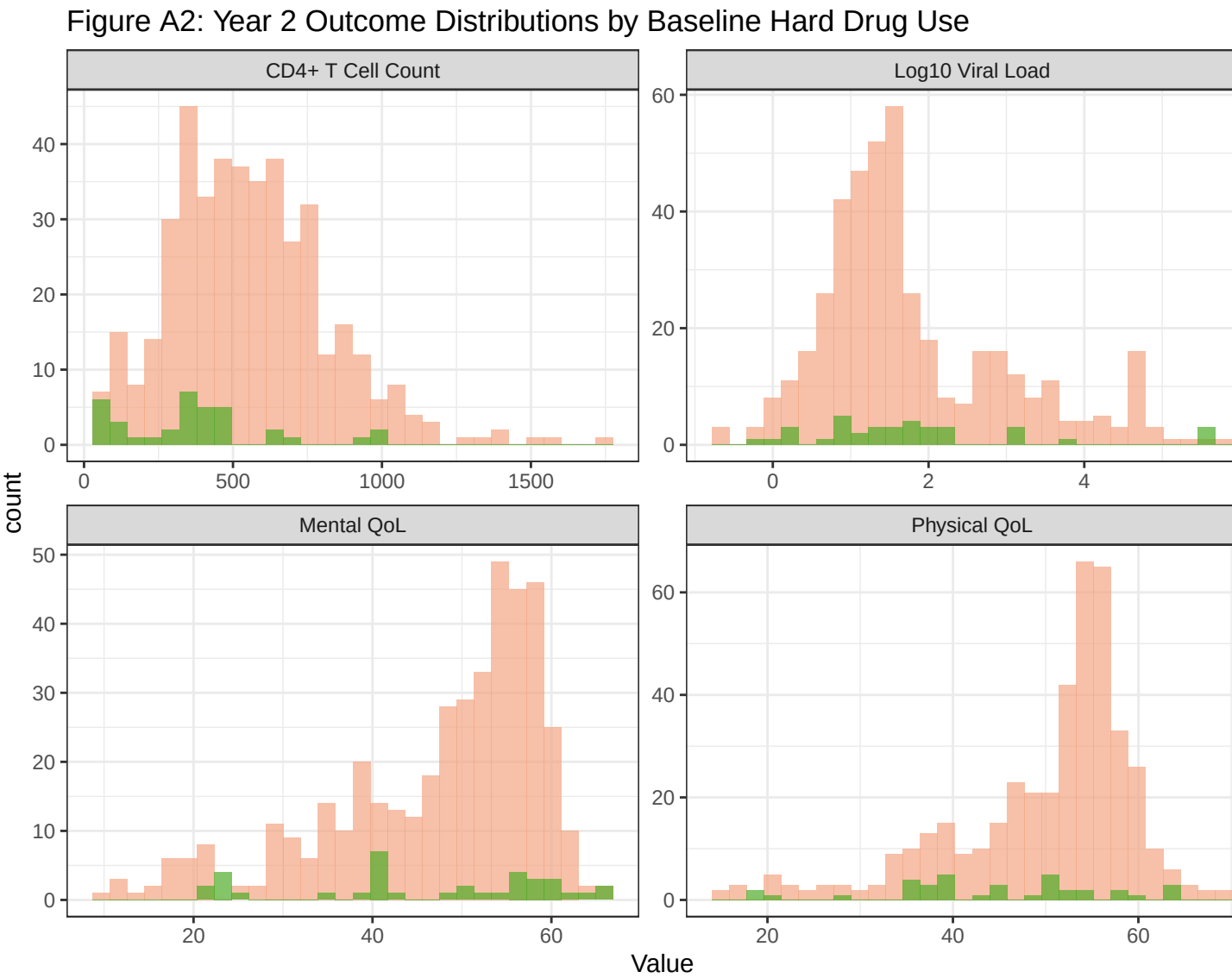
```

6.3 Figure A1: Longitudinal Outcome Trajectories

Figure A1: Mean Treatment Response Trajectories by Baseline Hard Drug Use

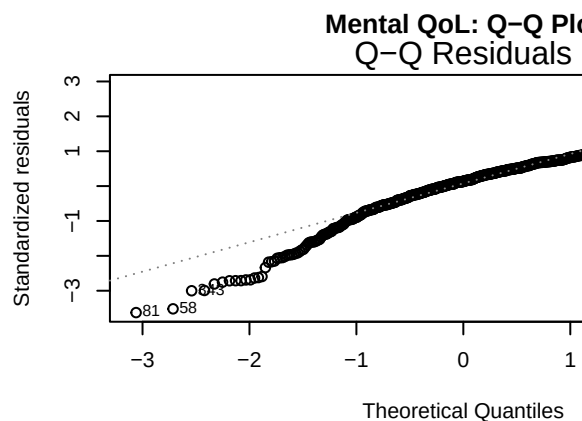
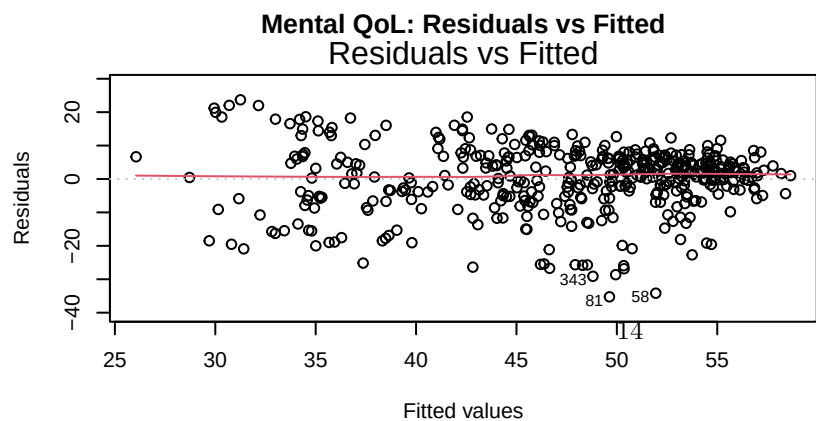
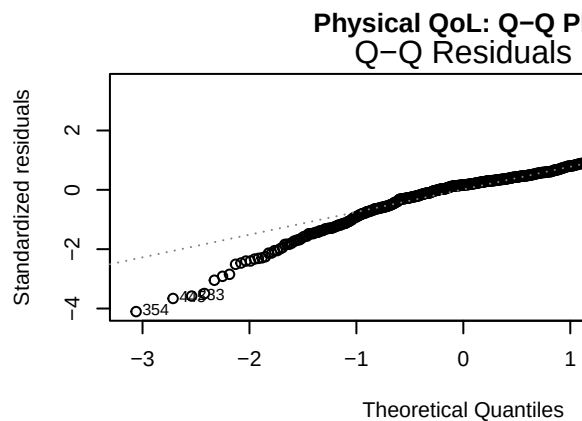
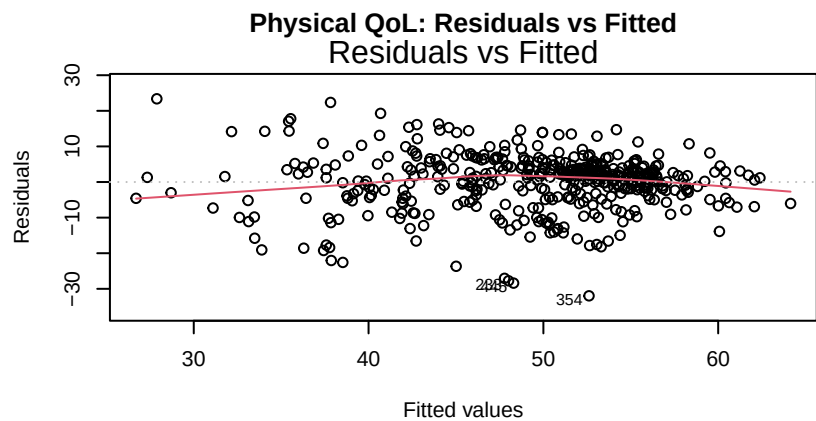
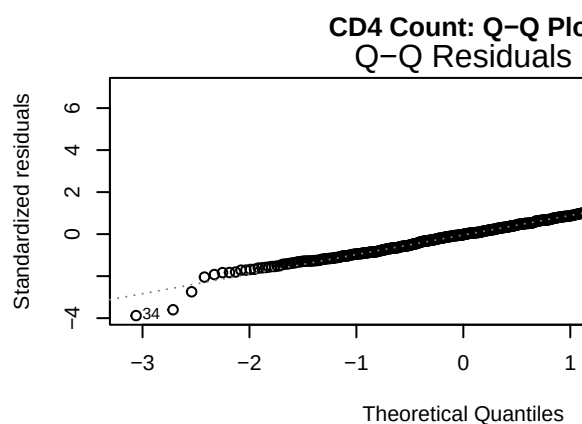
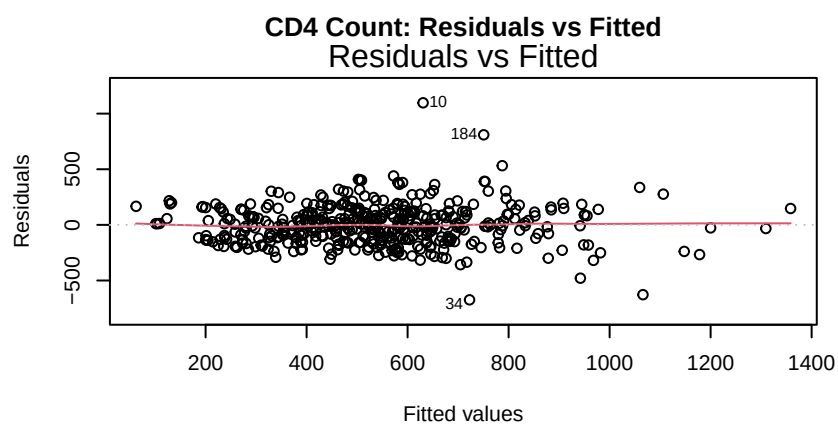
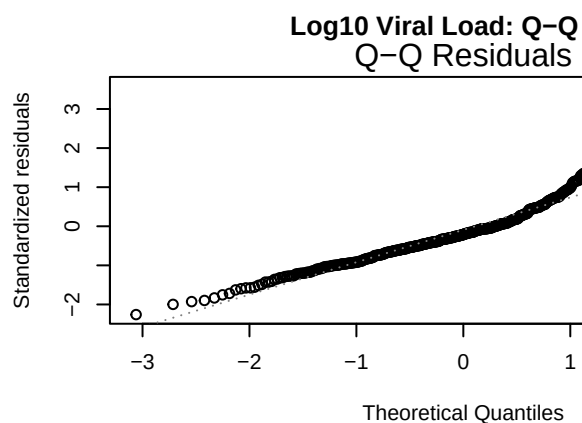
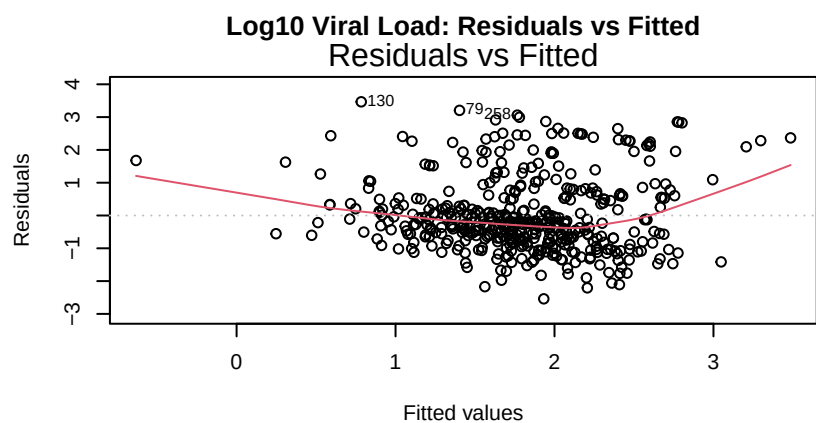


6.4 Figure A2: Year 2 Outcome Distributions



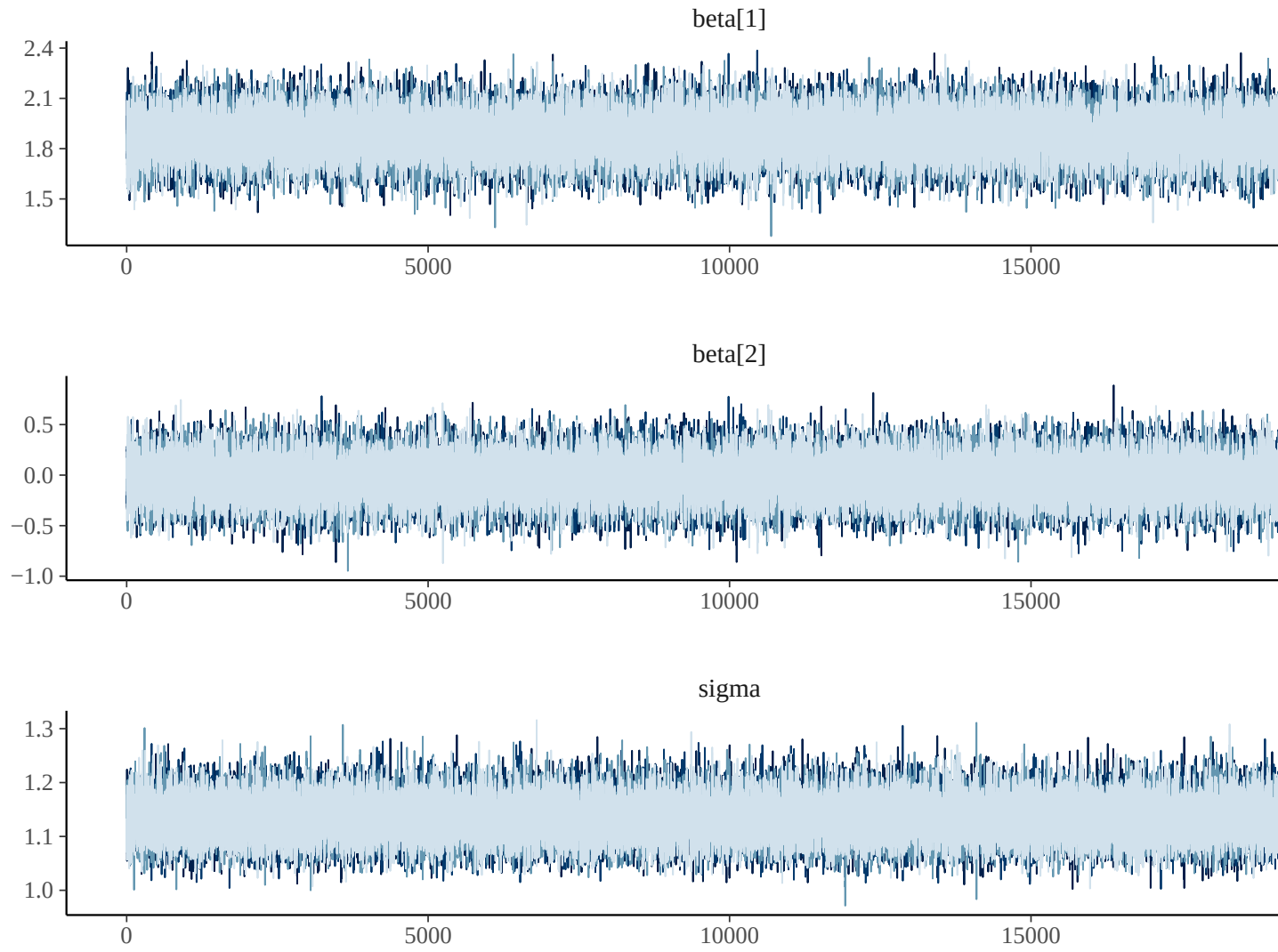
6.5 Figure A3: Frequentist Model Diagnostics

```
## pdf
## 2
```

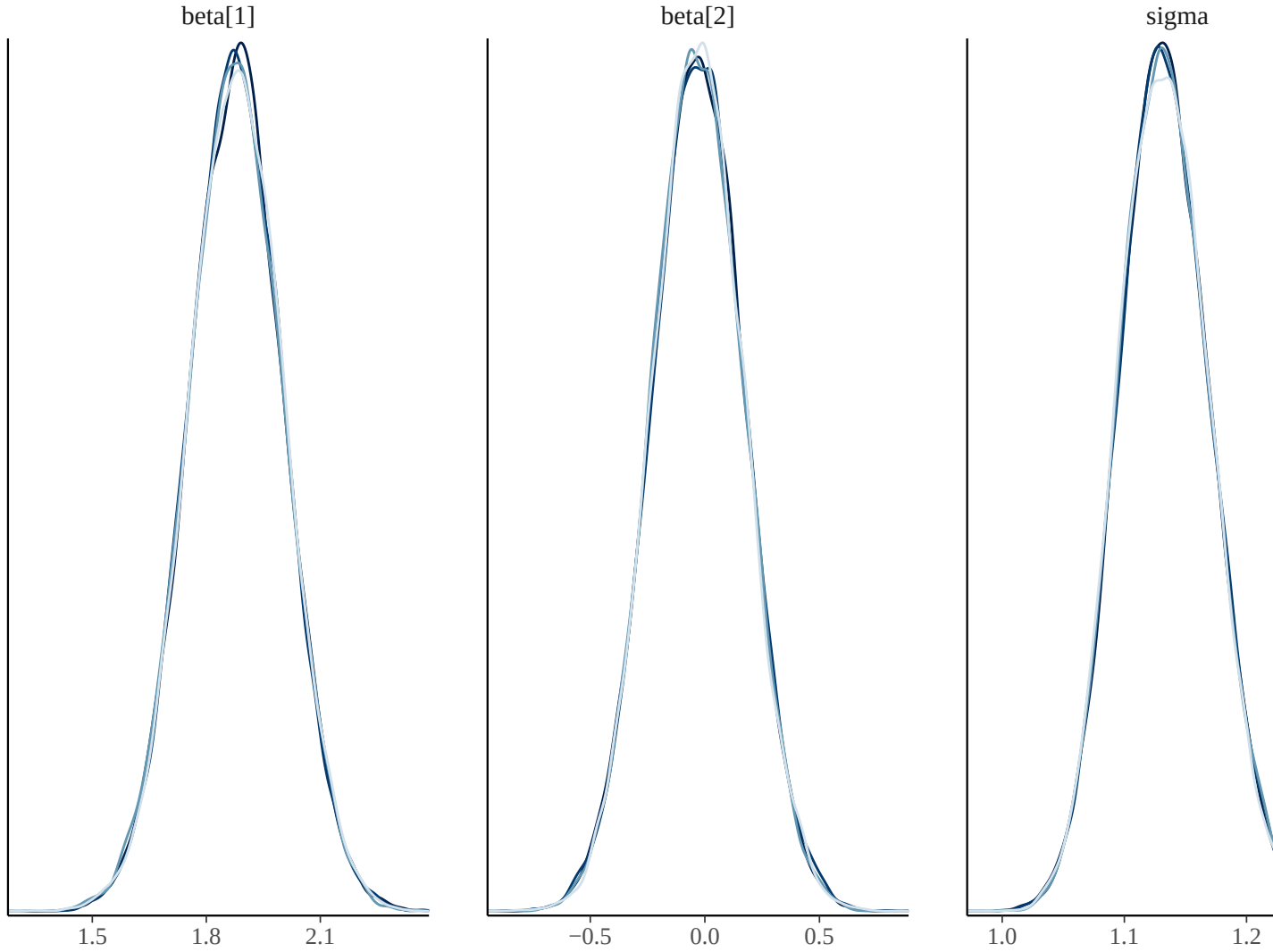


6.6 Figure A4: Bayesian MCMC Diagnostics (Log10 Viral Load, Vague Prior)

Trace Plots: Log10 Viral Load (Vague Prior)



Posterior Density Overlays: Log10 Viral Load (Vague Prior)



6.7 Table A3: Convergence Diagnostics

Table 14: Table A3: Convergence Diagnostics — Log10 Viral Load (Vague Prior)

variable	mean	sd	rhat	ess_bulk	ess_tail
beta[1]	1.8792	0.1258	1.0001	56376.99	57153.54
beta[2]	-0.0292	0.2021	1.0000	107354.41	59094.97
beta[3]	0.4454	0.0545	1.0002	111825.15	61114.06
beta[4]	-0.0211	0.0544	1.0000	103721.93	60684.54
beta[5]	0.2375	0.1164	1.0000	93877.89	63632.59
beta[6]	-0.1006	0.0554	1.0001	112442.74	60192.43
beta[7]	-0.1358	0.1144	1.0001	83952.39	62270.39
beta[8]	-0.2237	0.1364	1.0000	71706.79	63462.34

beta[9]	-0.0978	0.1371	1.0000	67001.23	63763.53
sigma	1.1338	0.0378	1.0000	107306.65	57656.84

6.8 Full Code

```
knitr::opts_chunk$set(
  echo = FALSE,
  warning = FALSE,
  message = FALSE,
  fig.width = 9,
  fig.height = 6,
  fig.align = "center"
)
library(tidyverse)
library(tableone)
library(knitr)
library(kableExtra)
library(patchwork)
library(cmdstanr)
library(posterior)
library(bayesplot)
library(loo)
library(HDIInterval)
proj_root <- "C:/Users/yunji/Desktop/College/CU Anschutz/2026 Spring/Advanced stats/BIOS6624/Project_1"
tables_dir <- file.path(proj_root, "Tables")
figs_dir <- file.path(proj_root, "Figures")
dir.create(tables_dir, showWarnings = FALSE)
dir.create(figs_dir, showWarnings = FALSE)

# Helper: save a ggplot to Figures/
save_fig <- function(plot, filename, width = 9, height = 6) {
  ggsave(file.path(figs_dir, filename), plot = plot,
    width = width, height = height, dpi = 300)
}

# Helper: save a kable as .tex to Tables/
save_tex <- function(kable_obj, filename) {
  tex <- kable(attr(kable_obj, "original_df"),
    format = "latex",
    booktabs = TRUE,
    caption = attr(kable_obj, "caption")) %>%
    kable_styling(latex_options = c("hold_position", "striped"))
  writeLines(as.character(tex), file.path(tables_dir, filename))
}

dat_raw <- read.csv(
  file.path(proj_root, "Raw_data/hiv_6624_final (3).csv"),
  stringsAsFactors = FALSE
)
dat <- dat_raw %>%
  mutate(
    AGG_PHYS = ifelse(AGG_PHYS %in% c(-1, 9, 99, 999), NA, AGG_PHYS),
    AGG_MENT = ifelse(AGG_MENT %in% c(-1, 9, 99, 999), NA, AGG_MENT),
```

Table 13: Table A3: Convergence Diagnostics — Log10 Viral Load (Vague Prior)

variable	mean	sd	rhat	ess_bulk	ess_tail
beta[1]	1.8792	0.1258	1.0001	56376.99	57153.54
beta[2]	-0.0292	0.2021	1.0000	107354.41	59094.97
beta[3]	0.4454	0.0545	1.0002	111825.15	61114.06
beta[4]	-0.0211	0.0544	1.0000	103721.93	60684.54
beta[5]	0.2375	0.1164	1.0000	93877.89	63632.59
beta[6]	-0.1006	0.0554	1.0001	112442.74	60192.43
beta[7]	-0.1358	0.1144	1.0001	83952.39	62270.39
beta[8]	-0.2237	0.1364	1.0000	71706.79	63462.34
beta[9]	-0.0978	0.1371	1.0000	67001.23	63763.53
sigma	1.1338	0.0378	1.0000	107306.65	57656.84

```

VLOAD = ifelse(VLOAD %in% c(-1, 999999999), NA, VLOAD),
LEU3N = ifelse(LEU3N == -1, NA, LEU3N),
CESD = ifelse(CESD == -1, NA, CESD),
BMI = ifelse(BMI < 10 | BMI > 70, NA, BMI),
ADH = ifelse(ADH == 9, NA, ADH),
income = ifelse(income == 9, NA, income),
EDUCBAS = ifelse(EDUCBAS == 9, NA, EDUCBAS),
DKGRP = ifelse(DKGRP == 9, NA, DKGRP),
SMOKE = ifelse(SMOKE == 9, NA, SMOKE)
) %>%
mutate(
  log10VLOAD = log10(VLOAD),
  race_cat = factor(ifelse(RACE == 1, "White", "Other"),
    levels = c("White", "Other")),
  educ_cat = factor(ifelse(EDUCBAS >= 5, "College degree+", "No college degree"),
    levels = c("No college degree", "College degree+")),
  smoke_cat = factor(case_when(SMOKE == 1 ~ "Never",
    SMOKE == 2 ~ "Former",
    SMOKE == 3 ~ "Current",
    TRUE ~ NA_character_),
    levels = c("Never", "Former", "Current")),
  adh_cat = factor(ADH, levels = 1:4,
    labels = c("100%", "95-99%", "75-94%", "<75%"),
    ordered = TRUE),
  hard_drugs_f = factor(hard_drugs, levels = c(0, 1), labels = c("No", "Yes"))
)

b1 <- dat %>%
  filter(years == 0, hivpos == 1) %>%
  select(newid, hard_drugs_f,
    log10VLOAD_b1 = log10VLOAD, LEU3N_b1 = LEU3N,
    AGG_PHYS_b1 = AGG_PHYS, AGG_MENT_b1 = AGG_MENT,
    age_b1 = age, BMI_b1 = BMI,
    race_cat, educ_cat, smoke_cat,
    CESD_b1 = CESD, DKGRP_b1 = DKGRP, income_b1 = income)

y2 <- dat %>%

```

```

filter(years == 2) %>%
select(newid,
       log10VLOAD_y2 = log10VLOAD, LEU3N_y2 = LEU3N,
       AGG_PHYS_y2 = AGG_PHYS,    AGG_MENT_y2 = AGG_MENT,
       ADH_y2 = ADH, adh_cat_y2 = adh_cat)

analytic_full <- inner_join(b1, y2, by = "newid") %>%
  filter(!is.na(hard_drugs_f))

analytic <- analytic_full %>%
  filter(
    !is.na(log10VLOAD_y2), !is.na(LEU3N_y2),
    !is.na(AGG_PHYS_y2),  !is.na(AGG_MENT_y2),
    !is.na(log10VLOAD_b1), !is.na(LEU3N_b1),
    !is.na(AGG_PHYS_b1),  !is.na(AGG_MENT_b1),
    !is.na(age_b1),       !is.na(race_cat),
    !is.na(BMI_b1),       !is.na(educ_cat), !is.na(smoke_cat)
  )
n_total <- nrow(analytic)
n_yes   <- sum(analytic$hard_drugs_f == "Yes")
n_no    <- sum(analytic$hard_drugs_f == "No")
fit_vl <- lm(
  log10VLOAD_y2 ~ hard_drugs_f + log10VLOAD_b1 + age_b1 + race_cat +
    BMI_b1 + educ_cat + smoke_cat + CESD_b1 + ADH_y2,
  data = analytic
)

fit_cd4 <- lm(
  LEU3N_y2 ~ hard_drugs_f + LEU3N_b1 + age_b1 + race_cat +
    BMI_b1 + educ_cat + smoke_cat + CESD_b1 + ADH_y2,
  data = analytic
)

fit_phys <- lm(
  AGG_PHYS_y2 ~ hard_drugs_f + AGG_PHYS_b1 + age_b1 + race_cat +
    BMI_b1 + educ_cat + smoke_cat + CESD_b1,
  data = analytic
)

fit_ment <- lm(
  AGG_MENT_y2 ~ hard_drugs_f + AGG_MENT_b1 + age_b1 + race_cat +
    BMI_b1 + educ_cat + smoke_cat + CESD_b1,
  data = analytic
)

extract_hd_coef <- function(fit, outcome_label) {
  cf <- summary(fit)$coefficients
  ci <- confint(fit)["hard_drugs_fYes", ]
  data.frame(
    Outcome = outcome_label,
    Estimate = round(cf["hard_drugs_fYes", 1], 4),
    SE       = round(cf["hard_drugs_fYes", 2], 4),
    p_value  = round(cf["hard_drugs_fYes", 4], 4),
  )
}

```

```

    CI_lower = round(ci[1], 4),
    CI_upper = round(ci[2], 4)
  )
}

freq_results <- bind_rows(
  extract_hd_coef(fit_vl, "Log10 Viral Load"),
  extract_hd_coef(fit_cd4, "CD4+ T Cell Count"),
  extract_hd_coef(fit_phys, "Physical QoL"),
  extract_hd_coef(fit_ment, "Mental QoL")
)

# Save .tex
kable(freq_results, format = "latex", booktabs = TRUE, row.names = FALSE,
      caption = "Adjusted Effect of Baseline Hard Drug Use on Year 2 Outcomes") %>%
  kable_styling(latex_options = c("hold_position", "striped")) %>%
  add_header_above(c(" " = 1, "Hard Drug Use vs. None (Adjusted)" = 5)) %>%
  { writeLines(as.character(.), file.path(tables_dir, "table_freq_results.tex")); . }

# Display in document
kable(freq_results, row.names = FALSE,
      caption = "Adjusted Effect of Baseline Hard Drug Use on Year 2 Outcomes") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE) %>%
  add_header_above(c(" " = 1, "Hard Drug Use vs. None (Adjusted)" = 5))

adh_tab <- analytic %>%
  filter(!is.na(ADH_y2)) %>%
  count(hard_drugs_f, ADH_y2) %>%
  group_by(hard_drugs_f) %>%
  mutate(
    pct = round(100 * n / sum(n), 1),
    ADH_label = factor(ADH_y2, levels = 1:4,
                      labels = c("100%", "95-99%", "75-94%", "<75%"))
  )

fig_adh <- ggplot(adh_tab, aes(x = ADH_label, y = pct, fill = hard_drugs_f)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("No" = "#F4A582", "Yes" = "#4DAC26")) +
  labs(title = "Figure 1: Year 2 ART Adherence by Baseline Hard Drug Use",
       x = "Adherence Level", y = "Percentage (%)",
       fill = "Baseline Hard\nDrug Use") +
  theme_bw()

save_fig(fig_adh, "fig1_adherence_by_drug_use.png")
fig_adh

chisq_adh <- chisq.test(table(analytic$hard_drugs_f, analytic$ADH_y2))
mediation_check <- function(fit_without, label) {
  df_adh <- analytic %>% filter(!is.na(ADH_y2))
  f_base <- update(fit_without, data = df_adh)
  f_adh <- update(f_base, . ~ . + ADH_y2)
  b0 <- coef(f_base)["hard_drugs_fYes"]
  b1 <- coef(f_adh)["hard_drugs_fYes"]
  pct <- 100 * (b0 - b1) / abs(b0)
  data.frame(

```

```

    Outcome           = label,
    Beta_No_Adherence  = round(b0, 4),
    Beta_With_Adherence = round(b1, 4),
    Pct_Change         = round(pct, 1),
    Adherence_Explains = ifelse(abs(pct) > 10, "Yes (>10%)", "No")
  )
}

med_results <- bind_rows(
  mediation_check(fit_vl, "Log10 Viral Load"),
  mediation_check(fit_cd4, "CD4+ T Cell Count"),
  mediation_check(fit_phys, "Physical QoL"),
  mediation_check(fit_ment, "Mental QoL")
)

kable(med_results, format = "latex", booktabs = TRUE, row.names = FALSE,
      caption = "Hard Drug Use Coefficient Before and After Adjusting for ART Adherence") %>%
  kable_styling(latex_options = c("hold_position", "striped")) %>%
  { writelines(as.character(.), file.path(tables_dir, "table_adherence_mediation.tex")); . }

kable(med_results, row.names = FALSE,
      caption = "Hard Drug Use Coefficient Before and After Adjusting for ART Adherence") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)

stan_code <- '
data {
  int<lower=0> N;
  int<lower=0> K;
  matrix[N, K] X;
  vector[N] y;
  vector[K] beta_prior_mean;
  vector<lower=0>[K] beta_prior_sd;
  real<lower=0> sigma_prior_sd;
}
parameters {
  vector[K] beta;
  real<lower=0> sigma;
}
model {
  beta ~ normal(beta_prior_mean, beta_prior_sd);
  sigma ~ normal(0, sigma_prior_sd);
  y ~ normal(X * beta, sigma);
}
generated quantities {
  vector[N] log_lik;
  for (n in 1:N)
    log_lik[n] = normal_lpdf(y[n] | X[n] * beta, sigma);
}
'

mod <- cmdstan_model(write_stan_file(stan_code))
build_stan_data <- function(outcome_var, bl_var, prior_sd_hd,
                           df = analytic, vague_sd = 100) {
  df_cc <- df %>%
    filter(complete.cases(.data[[outcome_var]], .data[[bl_var]]),

```

```

        hard_drugs_f, age_b1, race_cat, BMI_b1,
        educ_cat, smoke_cat)) %>%

mutate(
  hd_num      = as.numeric(hard_drugs_f == "Yes"),
  age_c       = scale(age_b1)[, 1],
  bl_c        = scale(.data[[bl_var]])[, 1],
  bmi_c       = scale(BMI_b1)[, 1],
  race_other  = as.numeric(race_cat == "Other"),
  educ_hi     = as.numeric(educ_cat == "College degree+"),
  smoke_form  = as.numeric(smoke_cat == "Former"),
  smoke_curr  = as.numeric(smoke_cat == "Current")
)
X      <- model.matrix(~ hd_num + bl_c + age_c + race_other +
                      bmi_c + educ_hi + smoke_form + smoke_curr,
                      data = df_cc)
y      <- df_cc[[outcome_var]]
K      <- ncol(X)
prior_sd <- rep(vague_sd, K)
prior_sd[2] <- prior_sd_hd
list(
  stan_data = list(N = nrow(X), K = K, X = X, y = y,
                  beta_prior_mean = rep(0, K),
                  beta_prior_sd   = prior_sd,
                  sigma_prior_sd  = vague_sd),

  df_cc = df_cc
)
}

summarize_posterior <- function(fit, gamma, outcome_label, prior_label) {
  draws <- as_draws_df(fit$draws())
  b     <- draws$`beta[2]`
  hpdi  <- hdi(b, credMass = 0.95)
  data.frame(
    Outcome      = outcome_label,
    Prior        = prior_label,
    Post_Mean    = round(mean(b), 4),
    Post_SD      = round(sd(b), 4),
    HPDI_Lower   = round(hpdi[1], 4),
    HPDI_Upper   = round(hpdi[2], 4),
    P_meaningful = round(mean(abs(b) > gamma), 4)
  )
}

gammas <- list(
  log10VLOAD_y2 = 0.5,
  LEU3N_y2      = 50,
  AGG_PHYS_y2   = 2,
  AGG_MENT_y2   = 2
)

outcomes <- list(
  list(y = "log10VLOAD_y2", bl = "log10VLOAD_b1", label = "Log10 Viral Load"),
  list(y = "LEU3N_y2",      bl = "LEU3N_b1",      label = "CD4+ T Cell Count"),
  list(y = "AGG_PHYS_y2",   bl = "AGG_PHYS_b1",   label = "Physical QoL"),

```

```

  list(y = "AGG_MENT_y2", bl = "AGG_MENT_bl", label = "Mental QoL")
)

bayes_results <- list()
fit_store <- list()

for (o in outcomes) {
  for (prior_type in c("Vague", "Skeptical")) {
    hd_sd <- if (prior_type == "Vague") 100 else 10
    sd_obj <- build_stan_data(o$y, o$bl, prior_sd_hd = hd_sd)
    fit <- mod$sample(
      data = sd_obj$stan_data,
      chains = 4,
      parallel_chains = 4,
      iter_sampling = 20000,
      iter_warmup = 5000,
      adapt_delta = 0.90,
      seed = 42,
      refresh = 0
    )
    key <- paste0(o$label, "_", prior_type)
    fit_store[[key]] <- fit
    bayes_results[[key]] <- summarize_posterior(
      fit, gamma = gammas[[o$y]], o$label, prior_type
    )
  }
}

post_tab <- bind_rows(bayes_results)
rownames(post_tab) <- NULL

kable(post_tab, format = "latex", booktabs = TRUE, row.names = FALSE,
  caption = "Bayesian Posterior Summaries for Hard Drug Use Coefficient",
  col.names = c("Outcome", "Prior", "Post. Mean", "Post. SD",
    "95% HPDI Low", "95% HPDI High", "P(|b|>g | data)")) %>%
  kable_styling(latex_options = c("hold_position", "striped")) %>%
  collapse_rows(columns = 1) %>%
  { writelines(as.character(.), file.path(tables_dir, "table_bayesian_posterior.tex")); . }

kable(post_tab, row.names = FALSE,
  caption = "Bayesian Posterior Summaries for Hard Drug Use Coefficient",
  col.names = c("Outcome", "Prior", "Post. Mean", "Post. SD",
    "95% HPDI Low", "95% HPDI High", "P(| |> | data)")) %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE) %>%
  collapse_rows(columns = 1)
loo_compare_results <- list()

for (o in outcomes) {
  sd_obj <- build_stan_data(o$y, o$bl, prior_sd_hd = 100)
  df_cc <- sd_obj$df_cc

  X_null <- model.matrix(~ bl_c + age_c + race_other + bmi_c +
    educ_hi + smoke_form + smoke_curr, data = df_cc)
  y_vec <- df_cc[[o$y]]

```

```

K_null <- ncol(X_null)

fit_null <- mod$sample(
  data = list(N = nrow(X_null), K = K_null, X = X_null, y = y_vec,
    beta_prior_mean = rep(0, K_null),
    beta_prior_sd = rep(100, K_null),
    sigma_prior_sd = 100),
  chains = 4, parallel_chains = 4,
  iter_sampling = 20000, iter_warmup = 5000,
  seed = 42, refresh = 0
)

ll_full <- as_draws_matrix(fit_store[[paste0(o$label, "_Vague")]]$draws("log_lik"))
ll_null <- as_draws_matrix(fit_null$draws("log_lik"))
loo_full <- loo(ll_full)
loo_null_res <- loo(ll_null)
comp <- loo_compare(loo_null_res, loo_full)

loo_compare_results[[o$label]] <- data.frame(
  Outcome = o$label,
  LOO_Full = round(loo_full$estimates["elpd_loo", "Estimate"], 2),
  LOO_Null = round(loo_null_res$estimates["elpd_loo", "Estimate"], 2),
  Delta_LOO = round(diff(comp[, "elpd_diff"]), 2),
  SE_Delta = round(diff(comp[, "se_diff"]), 2)
)
}

loo_tab <- bind_rows(loo_compare_results)

kable(loo_tab, format = "latex", booktabs = TRUE, row.names = FALSE,
  caption = "LOO-IC Model Comparison: Full vs. Null Model (Without Hard Drug Use). Delta LOO > 3 fa",
  kable_styling(latex_options = c("hold_position", "striped")) %>%
  { writelines(as.character(.), file.path(templates_dir, "table_loo_comparison.tex")); . }

kable(loo_tab, row.names = FALSE,
  caption = "LOO-IC Model Comparison: Full vs. Null Model (Without Hard Drug Use). Delta LOO > 3 fa",
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
freq_comp <- freq_results %>%
  mutate(Method = "Frequentist (95% CI)") %>%
  select(Outcome, Method, Est = Estimate, Low = CI_lower, High = CI_upper)

bayes_comp <- post_tab %>%
  mutate(Method = paste0("Bayesian ", Prior, " (95% HPDI)") %>%
  select(Outcome, Method, Est = Post_Mean, Low = HPDI_Lower, High = HPDI_Upper)

comparison_tab <- bind_rows(freq_comp, bayes_comp) %>%
  mutate(Method = factor(Method, levels = c(
    "Frequentist (95% CI)",
    "Bayesian Vague (95% HPDI)",
    "Bayesian Skeptical (95% HPDI)"
  )))

fig_compare <- ggplot(comparison_tab,

```

```

    aes(x = Est, xmin = Low, xmax = High, y = Method, color = Method)) +
  geom_pointrange(size = 0.7) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  facet_wrap(~ Outcome, scales = "free_x", ncol = 2) +
  scale_color_manual(values = c("#2166AC", "#D6604D", "#4DAC26")) +
  labs(title = "Figure 2: Frequentist and Bayesian Estimates for Hard Drug Use Effect",
        subtitle = "Point estimates with 95% CI / HPDI",
        x = "Estimate", y = NULL, color = NULL) +
  theme_bw() +
  theme(legend.position = "bottom", axis.text.y = element_blank())

save_fig(fig_compare, "fig2_freq_bayes_comparison.png", width = 9, height = 5)
fig_compare
sens_tab <- post_tab %>%
  select(Outcome, Prior, Post_Mean, Post_SD, HPDI_Lower, HPDI_Upper, P_meaningful)

kable(sens_tab, format = "latex", booktabs = TRUE, row.names = FALSE,
      caption = "Prior Sensitivity: Vague vs. Skeptical Priors",
      col.names = c("Outcome", "Prior", "Post. Mean", "Post. SD",
                    "HPDI Lower", "HPDI Upper", "P(|b|>g | data)")) %>%
  kable_styling(latex_options = c("hold_position", "striped")) %>%
  collapse_rows(columns = 1) %>%
  { writeLines(as.character(.), file.path(tables_dir, "table_prior_sensitivity.tex")); . }

kable(sens_tab, row.names = FALSE,
      caption = "Prior Sensitivity: Vague vs. Skeptical Priors",
      col.names = c("Outcome", "Prior", "Post. Mean", "Post. SD",
                    "HPDI Lower", "HPDI Upper", "P(| |> | data)")) %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE) %>%
  collapse_rows(columns = 1)
vars_check <- c("log10VLOAD_y2", "LEU3N_y2", "AGG_PHYS_y2", "AGG_MENT_y2",
               "log10VLOAD_b1", "LEU3N_b1", "AGG_PHYS_b1", "AGG_MENT_b1",
               "age_b1", "BMI_b1", "race_cat", "educ_cat", "smoke_cat",
               "CESD_b1", "ADH_y2")

miss_tab <- data.frame(
  Variable = vars_check,
  N_Missing = sapply(vars_check, function(v) sum(is.na(analytic_full[[v]]))),
  Pct_Missing = sapply(vars_check, function(v)
    round(100 * mean(is.na(analytic_full[[v]])), 1))
)

kable(miss_tab, format = "latex", booktabs = TRUE, row.names = FALSE,
      caption = "Table A1: Missing Data Summary") %>%
  kable_styling(latex_options = c("hold_position", "striped")) %>%
  { writeLines(as.character(.), file.path(tables_dir, "tableA1_missing_data.tex")); . }

kable(miss_tab, row.names = FALSE, caption = "Table A1: Missing Data Summary") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
tab1 <- CreateTableOne(
  vars = c("age_b1", "race_cat", "educ_cat", "BMI_b1", "smoke_cat",
           "CESD_b1", "DKGRP_b1", "log10VLOAD_b1", "LEU3N_b1",
           "AGG_PHYS_b1", "AGG_MENT_b1"),

```

```

strata      = "hard_drugs_f",
data        = analytic,
factorVars  = c("race_cat", "educ_cat", "smoke_cat"),
addOverall  = TRUE
)
print(tab1, nonnormal = c("CESD_b1", "log10VLOAD_b1", "LEU3N_b1", "DKGRP_b1"),
      showAllLevels = TRUE)
traj <- dat %>%
  filter(hivpos == 1, years %in% 0:8) %>%
  select(-any_of("hard_drugs_f")) %>%
  inner_join(b1 %>% select(newid, hard_drugs_f), by = "newid") %>%
  filter(!is.na(hard_drugs_f)) %>%
  group_by(years, hard_drugs_f) %>%
  summarise(
    mean_VL   = mean(log10VLOAD, na.rm = TRUE),
    mean_CD4  = mean(LEU3N,      na.rm = TRUE),
    mean_PHYS = mean(AGG_PHYS,   na.rm = TRUE),
    mean_MENT = mean(AGG_MENT,   na.rm = TRUE),
    .groups   = "drop"
  )

p1 <- ggplot(traj, aes(years, mean_VL, color = hard_drugs_f, group = hard_drugs_f)) +
  geom_line(size = 1) + geom_point(size = 2) +
  labs(title = "Log10 Viral Load", x = "Years since HAART", y = "Mean Log10 VL",
        color = "Baseline\nHard Drug Use") + theme_bw()

p2 <- ggplot(traj, aes(years, mean_CD4, color = hard_drugs_f, group = hard_drugs_f)) +
  geom_line(size = 1) + geom_point(size = 2) +
  labs(title = "CD4+ T Cell Count", x = "Years since HAART", y = "Mean CD4 (cells/mL)",
        color = "Baseline\nHard Drug Use") + theme_bw()

p3 <- ggplot(traj, aes(years, mean_PHYS, color = hard_drugs_f, group = hard_drugs_f)) +
  geom_line(size = 1) + geom_point(size = 2) +
  labs(title = "Physical QoL (SF-36)", x = "Years since HAART", y = "Mean Score",
        color = "Baseline\nHard Drug Use") + theme_bw()

p4 <- ggplot(traj, aes(years, mean_MENT, color = hard_drugs_f, group = hard_drugs_f)) +
  geom_line(size = 1) + geom_point(size = 2) +
  labs(title = "Mental QoL (SF-36)", x = "Years since HAART", y = "Mean Score",
        color = "Baseline\nHard Drug Use") + theme_bw()

fig_traj <- (p1 + p2) / (p3 + p4) +
  plot_annotation(title = "Figure A1: Mean Treatment Response Trajectories by Baseline Hard Drug Use")

save_fig(fig_traj, "figA1_trajectories.png", width = 9, height = 8)
fig_traj
fig_dist <- analytic %>%
  select(hard_drugs_f, log10VLOAD_y2, LEU3N_y2, AGG_PHYS_y2, AGG_MENT_y2) %>%
  pivot_longer(cols = -hard_drugs_f, names_to = "outcome", values_to = "value") %>%
  mutate(outcome = recode(outcome,
    "log10VLOAD_y2" = "Log10 Viral Load",
    "LEU3N_y2"      = "CD4+ T Cell Count",
    "AGG_PHYS_y2"   = "Physical QoL",

```

```

  "AGG_MENT_y2" = "Mental QoL")) %>%
  ggplot(aes(x = value, fill = hard_drugs_f)) +
  geom_histogram(alpha = 0.7, bins = 30, position = "identity") +
  facet_wrap(~ outcome, scales = "free") +
  scale_fill_manual(values = c("No" = "#F4A582", "Yes" = "#4DAC26")) +
  labs(title = "Figure A2: Year 2 Outcome Distributions by Baseline Hard Drug Use",
       x = "Value", fill = "Baseline Hard Drug Use") +
  theme_bw()

save_fig(fig_dist, "figA2_outcome_distributions.png", width = 9, height = 6)
fig_dist
png(file.path(figs_dir, "figA3_freq_diagnostics.png"),
    width = 9, height = 10, units = "in", res = 300)
par(mfrow = c(4, 2))
plot(fit_vl, which = 1, main = "Log10 Viral Load: Residuals vs Fitted")
plot(fit_vl, which = 2, main = "Log10 Viral Load: Q-Q Plot")
plot(fit_cd4, which = 1, main = "CD4 Count: Residuals vs Fitted")
plot(fit_cd4, which = 2, main = "CD4 Count: Q-Q Plot")
plot(fit_phys, which = 1, main = "Physical QoL: Residuals vs Fitted")
plot(fit_phys, which = 2, main = "Physical QoL: Q-Q Plot")
plot(fit_ment, which = 1, main = "Mental QoL: Residuals vs Fitted")
plot(fit_ment, which = 2, main = "Mental QoL: Q-Q Plot")
par(mfrow = c(1, 1))
dev.off()

par(mfrow = c(4, 2))
plot(fit_vl, which = 1, main = "Log10 Viral Load: Residuals vs Fitted")
plot(fit_vl, which = 2, main = "Log10 Viral Load: Q-Q Plot")
plot(fit_cd4, which = 1, main = "CD4 Count: Residuals vs Fitted")
plot(fit_cd4, which = 2, main = "CD4 Count: Q-Q Plot")
plot(fit_phys, which = 1, main = "Physical QoL: Residuals vs Fitted")
plot(fit_phys, which = 2, main = "Physical QoL: Q-Q Plot")
plot(fit_ment, which = 1, main = "Mental QoL: Residuals vs Fitted")
plot(fit_ment, which = 2, main = "Mental QoL: Q-Q Plot")
par(mfrow = c(1, 1))
fit_vl_vague <- fit_store[["Log10 Viral Load_Vague"]]

fig_trace <- mcmc_trace(fit_vl_vague$draws(), pars = c("beta[1]", "beta[2]", "sigma"),
  facet_args = list(ncol = 1)) +
  ggtitle("Trace Plots: Log10 Viral Load (Vague Prior)")

fig_dens <- mcmc_dens_overlay(fit_vl_vague$draws(),
  pars = c("beta[1]", "beta[2]", "sigma")) +
  ggtitle("Posterior Density Overlays: Log10 Viral Load (Vague Prior)")

save_fig(fig_trace, "figA4a_mcmc_trace.png", width = 9, height = 6)
save_fig(fig_dens, "figA4b_mcmc_density.png", width = 9, height = 6)

fig_trace
fig_dens
conv_tab <- summarise_draws(fit_vl_vague$draws()) %>%
  filter(grepl("beta|sigma", variable)) %>%
  select(variable, mean, sd, rhat, ess_bulk, ess_tail)

```

```

kable(conv_tab, format = "latex", booktabs = TRUE,
      caption = "Table A3: Convergence Diagnostics - Log10 Viral Load (Vague Prior)",
      digits = 4) %>%
kable_styling(latex_options = c("hold_position", "striped")) %>%
{ writeLines(as.character(.), file.path(tables_dir, "tableA3_convergence.tex")); . }

kable(conv_tab, digits = 4,
      caption = "Table A3: Convergence Diagnostics - Log10 Viral Load (Vague Prior)" %>%
kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)

```